	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering**	
NLP FA I : Case Study Report		

Semantic Plagiarism Detection for Academic Paper

1. Introduction:

Natural Language Processing (NLP) is a subfield of AI that focuses on enabling computers to understand, interpret and generate human language in a meaningful manner. Over the past few decades, NLP has been transformed achieving great results using deep learning architectures, transformer based models and many more. These developments have revolutionized various applications including machine translation, sentiment analysis, question answering, and document understanding.

Modern NLP systems leverage contextual word embeddings that capture semantic relationships between words and sentences. Unlike traditional bag-of-words or TF-IDF representations that treat text as independent tokens, contemporary approaches like BERT (Bidirectional Encoder Representations from Transformers) and its variants understand language in context, enabling a more nuanced understanding of meaning, syntax, and pragmatics.

1.1 Real word applications of NLP

NLP plays a crucial role in numerous real-world applications, including:

- Machine translation
- Chatbots and virtual assistants
- Sentiment analysis
- Text summarization
- Information retrieval
- Question answering systems
- Plagiarism detection

In academic and research environments, plagiarism detection has become increasingly important due to the rapid growth of digital content and easy access to online resources. Institutions require automated systems that can efficiently analyze large volumes of textual data to maintain academic integrity.

Modern plagiarism detection systems must go beyond exact word matching and be capable of identifying paraphrased or semantically similar content. This is where advanced NLP techniques become essential.

1.2 Motivation

With the tremendous availability of AI / GPT (Generated Pretrained Tools) for content generation and instance paraphrase, plagiarism has increased significantly .Students modify sentences and replace words with their synonyms or use humanizer tools to remove plagiarism.

To maintain the originality of the research work and the content to be published should be without any copied idea. Conventional Plagiarism detection technique focuses more on lexical similarity and fail to identify :

1) Semantic similarity
2) Content is paraphrased
3) Sentence has been altered (voice change , from active to passive voice)

This limitation motivated the selection of a semantic plagiarism detection system, which focuses on identifying similarity in meaning rather than exact word matches.

By leveraging modern embedding models such as Sentence-BERT (SBERT), it becomes possible to capture contextual and semantic information, thereby enabling more robust detection of paraphrased content.

1.3 Research Gap / Limitations in the Existing System

Existing plagiarism detection systems can broadly be categorized into two types: lexical-based systems and commercial proprietary systems. Lexical-based approaches, such as TF-IDF, n-gram analysis, and string matching, rely primarily on surface-level textual similarity. While these methods are computationally efficient and easy to implement, they are limited in their ability to detect semantic plagiarism, particularly in cases involving paraphrasing, synonym substitution, or structural modifications.

In contrast, commercial platforms such as Turnitin employ large-scale databases and proprietary algorithms to perform similarity analysis. However, these systems lack transparency, are not customizable for domain-specific academic datasets, and are generally inaccessible for research experimentation due to their closed architectures.

Moreover, many traditional systems fail to capture contextual meaning, struggle with semantically equivalent but lexically different content, and provide limited interpretability in similarity computation. Consequently, there exists a research gap in the development of an open, explainable, and semantic similarity-based plagiarism detection framework that leverages modern NLP techniques and operates effectively within a controlled academic dataset.

1.4 Contribution of the Proposed Work

This work proposes a hybrid hierarchical framework for semantic plagiarism detection in academic computer science papers by combining lexical and semantic similarity techniques.

The main contributions are summarized as follows:

- **Hybrid Lexical–Semantic Framework:**
  Integrates TF-IDF with Sentence-BERT embeddings to capture both surface-level text overlap and deeper contextual similarity, enabling robust detection of paraphrased plagiarism.
- **Two-Phase Coarse-to-Fine Architecture:**
  Phase-1 performs efficient document filtering using titles and abstracts, while Phase-2 applies detailed semantic analysis on full sections and sentences, reducing computation while preserving accuracy.
- **Hierarchical Plagiarism Analysis:**
  Introduces document-, section-, and sentence-level evaluation for precise localization of plagiarized content.
- **Section-Aware Weighted Scoring:**
  Assigns higher importance to critical sections such as methodology and results, improving intellectual plagiarism assessment.
- **Scalable Semantic Retrieval:**
  Uses FAISS-based indexing of SBERT embeddings for fast similarity search across large paper collections.
- **Explainable Detection Output:**
  Provides section-wise scores and highlighted sentences, supporting transparent and interpretable plagiarism reports.

Overall, the proposed system offers a scalable, explainable, and semantically grounded plagiarism detection approach that overcomes limitations of traditional keyword-based methods while remaining practical for academic deployment.

2. Problem Statement:

Traditional plagiarism detection systems rely primarily on lexical similarity techniques such as TF-IDF, n-grams, and string matching. These approaches fail to effectively detect semantic plagiarism, where the meaning of the content is preserved but the wording is altered through paraphrasing or structural changes.

With the increasing availability of digital academic resources and AI-assisted writing tools, detecting paraphrased plagiarism has become a critical challenge for educational institutions. Reliable and automated semantic plagiarism detection is essential to maintain academic integrity and ensure originality in research submissions.

Challenges Involved

Semantic plagiarism detection involves several challenges, including contextual ambiguity, variations in writing style, synonym substitution, structural reordering, and noise introduced during PDF text extraction. Capturing true semantic similarity while minimizing false positives remains a complex task.

Objective of the Proposed Solution

The objective for the work is to develop a system which detects

1) To detect semantic plagiarism in computer science research papers using a hybrid lexical–semantic approach (Sentence Bert and TF-IDF).
2) To incorporate section-aware weighted scoring to emphasize critical research components.
3) To enable scalable semantic retrieval using FAISS indexing.
4) To generate explainable plagiarism reports highlighting plagiarized sections and sentences.

3. Literature Review:

Title of the Research Paper (Year)	Methodology (Model used)	Key Finding	Research Gap / Limitation
An Intelligent Approach for Semantic Plagiarism Detection in Scientific Papers (2021)	**Clustering & DL:** Mini-Batch K-Means + Universal Sentence Encoder (USE)	Reliable accuracy comparable to Turnitin; effective for semantic detection	Slower execution than Turnitin; limited corpus size (500 docs)

PLAGISENSE: An AI-Based Semantic Plagiarism Detection System Using Sentence-BERT (2023)	**Transformer:** Sentence-BERT (SBERT) embeddings + Cosine Similarity	Detects paraphrasing (60-80% similarity) better than keyword matching	No multilingual support; lacks visual highlighting for users
A Novel Plagiarism Detection Approach Combining BERT-based Word Embedding, Attention-based LSTMs and an Improved Differential Evolution Algorithm (2024)	**BPD-IDE:** BERT embeddings + Attention Bi-LSTMs + Differential Evolution	BERT reduced error rates significantly	BERT is computationally heavy; plans to use lighter DistilBERT & RL
Beyond Black Box AI-Generated Plagiarism Detection: From Sentence to Document Level (2023)	**Hybrid NLP:** T5 paraphrasing + GPT-3.5 + Cosine Similarity + LDA	**94% accuracy** on HC3 dataset; provides sentence-level transparency	Paraphrasing did not improve results on GPT-wiki-intro; needs unbiased datasets
An effective text plagiarism detection system based on feature selection and SVM techniques (2023)	**SVM & Selection:** SVM classifier + Chi-square selection on 34 features	**92.91% Plag score** on PAN 2014; identified 32 key features	Lacks cross-lingual detection; needs training on diverse confusion types
Semantic Plagiarism Detection of Figures in Scholarly Documents: A Conceptual Framework (2023)	**Conceptual Framework:** ResNet-50 classification + OCR/Pytesseract + Jaccard/Edit Distance	Targets figure-based plagiarism using AUC and metadata	Framework only; parsing complex/cluttered figures remains difficult

4. Dataset Description:

The dataset for the proposed semantic detection would consists of approximately 300-400 research paper focusing on the domains of machine learning , data science , artificial Intelligence, software engineering, and other computer science related fields.

Each paper will be in the PDF format and further divided into different sections of research paper (Title , Abstract , Introduction , Methodology , Result and Conclusion) will be segmented. Sentences from the Dataset are embedded using Sentence - Bert and indexed for semantic retrieval using FAISS. Along with TF-IDF and cosine similarity to identify the lexical similarity .

Research Paper are collected from open access academic repositories including -

1) IEEE Xplore - engineering and computing journals
2) ACM Digital Library - computer science and journal papers
3) arXiv - widely used for CS preprints

5. Proposed Methodology:

This section describes the "Hybrid approach for Semantic Plagiarism Detection" mainly focused on academic research papers. The Hybrid approach detects both lexical as well as semantic plagiarism by combining traditional text similar checking method like TF-IDF, context embeddings using Sentence Bert and assigning section wise weighting, This methodology works in multistage pipeline , which includes document ingestion, preprocessing , hybrid feature representation ,similarity computation and weighted plagiarism scoring.

5.1 System Overview

A student research paper is the input provided to the system, the system automatically compares it with the pre-indexed research papers from the dataset to identify the semantic similarity. The approach is further divided into 3 levels:

a) Document Level - This helps to identify the set of similar research papers for the dataset.
b) Section Level - Comparison among sections of research paper.
    i) Abstract (student's paper) - Abstract (referenced research paper)
    ii) Methodology (student's paper) - Methodology(referenced research paper) etc
c) Sentence Level - This detects the exact plagiarized content .

This includes frameworks as:

1) TF-IDF to check Lexical similarity
2) Sentence BERT, built on BERT to check contextual semantic similarity using embedding generated.
3) FAISS for efficient semantic retrieval

5.2 Document Ingestion and Section Segmentation

As the data input and the referenced data is in the form of PDF/Docx format , it needs to be converted into raw text. Each document is segmented into different sections of research paper as
- Abstract, Introduction, Methodology, Result, Conclusion.

5.3 Text Preprocessing

Each section of the research paper undergoes text preprocessing which includes:

1) Lowercasing - Normalizing the text to remove ambiguity among capital and lowercase words.
2) Tokenization - Splitting of texts into words.
3) Stopword removal - Removal of words which don't change the semantic meaning.
4) Lemmatization - Convert the words into their base form.
5) Noise Removal - Removal of references, emojis and the equations in the paper.

These steps help to improve lexical and semantic representation of the data.

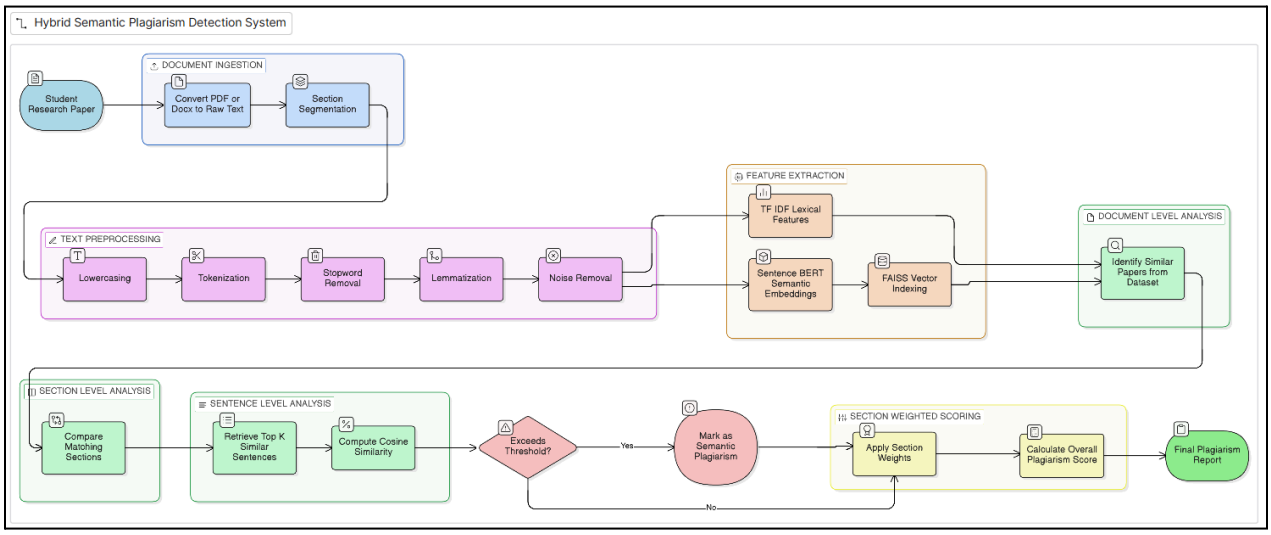
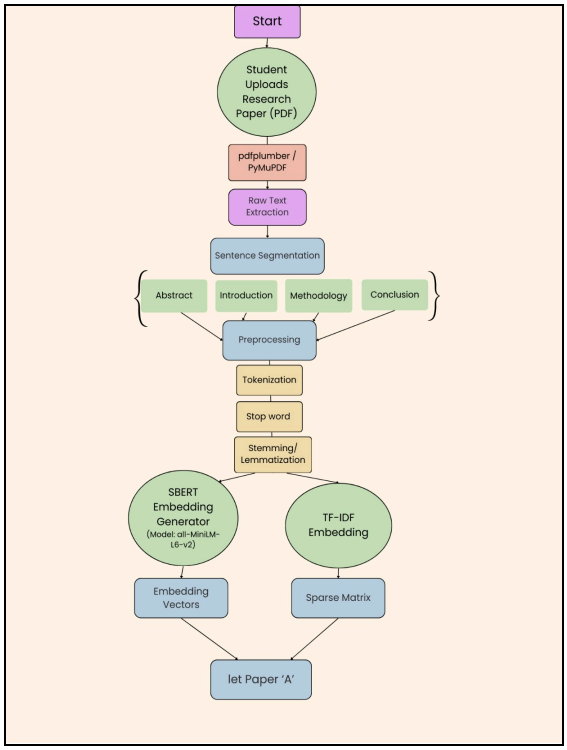
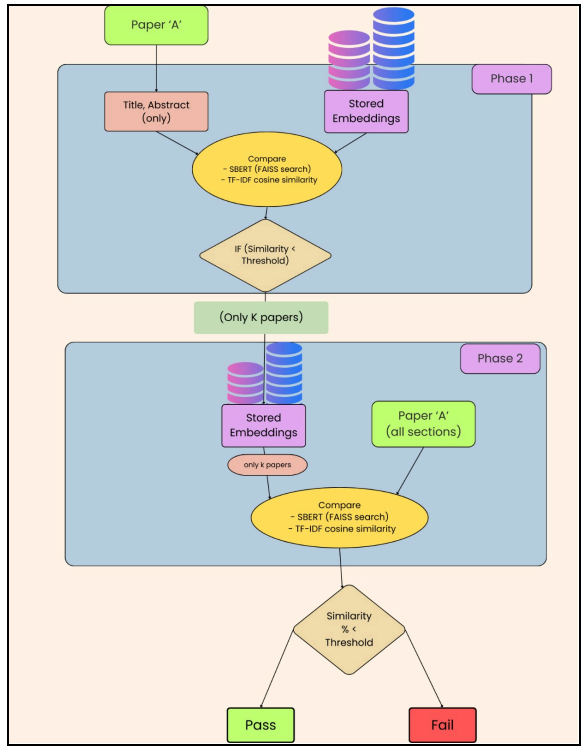


Fig.1 System Architecture of Hybrid Semantic Plagiarism Detection

5.4 Feature Representation

1) TF-IDF - Lexical Feature
   TF-IDF vectors are calculated for all the documents in the dataset to identify exact copy and duplicate text. This filters papers with high semantic matching to perform further deeper semantic analysis.
2) Sentence BERT - Semantic Features
   This is an enhanced version of BERT. Each sentence is encoded to dense vectors. SBERT generates sentence level embedding for cosine similarity, this helps to detect paraphrased and idea plagiarised content.
3) FAISS - Vectors Indexing and Retrieval
   All the vectors generated by SBERT are stored in the FAISS vector index, This makes it efficient to search over large datasets .
   FAISS helps to retrieve the top k most similar reference sentences and then the cosine similarity score is computed.





$$\text{CosineSimilarity(A,B)} = \frac{A.B}{\|A\| \times \|B\|}$$

If the similarity exceeds the predefined threshold , then the sentences are marked as semantic plagiarism.

5.5 Section Weighted Similarity

Different sections of the research paper differ in their originality. Therefore weights are predefined according to the sections of the research paper. (ie. Sections like methodology and result should have uniqueness, thus reducing the plagiarism content in the research paper must be assigned with high weight as compared to other sections in the research paper).

For example : Abstract - 0.15
Introduction - 0.15
Methodology - 0.35
Results - 0.25
Conclusion -0.10

The final plagiarism score can be calculated by

$$\text{Overall Plagiarism} = \sum_{i=1}^{n} (S_i * W_i)$$

This ensures that plagiarism in Methodology and results must be less to achieve an overall less plagiarism score. Thus Results and Methodology impact the final score more strongly than the plagiarism in the introductory section in the research paper.

6. Requirements for Implementation:

6.1 Programming Language and Libraries

1) Programming Language : Python.
2) PyMuPDF (fitz) : for pdf processing.
3) SpaCy : for lemmatization, section parsing and fast preprocessing.
4) NLTK : for removal of stop words.
5) Sckit-learn : Calculates cosine similarity , TF-IDF for lexical analysis.
6) Numpy : numerical processing and data handling.
7) Sentence Bert for generating sentence embedding.
8) FAISS : helps in fast retrieval of sentence embeddings.

7. References:

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